Weighted Weil-restriction decompositions

2026-10-19

Native larger-curve diagnostic

This run uses the repository's native cubic-field arithmetic and S3 resultant pair solver. It constructs no covering curve and no whole-field or whole-group operation table. Each decision is cross-checked with a separate signed-pair dictionary unavailable to the solver.

The local reference is the identical hybrid with c=1, not the best complete IC pipeline. Counts below are stage diagnostics, not end-to-end speedups. Complete S, ratios to rho and external fastest-tool comparisons are unestablished. Wall time is not admitted at L0.

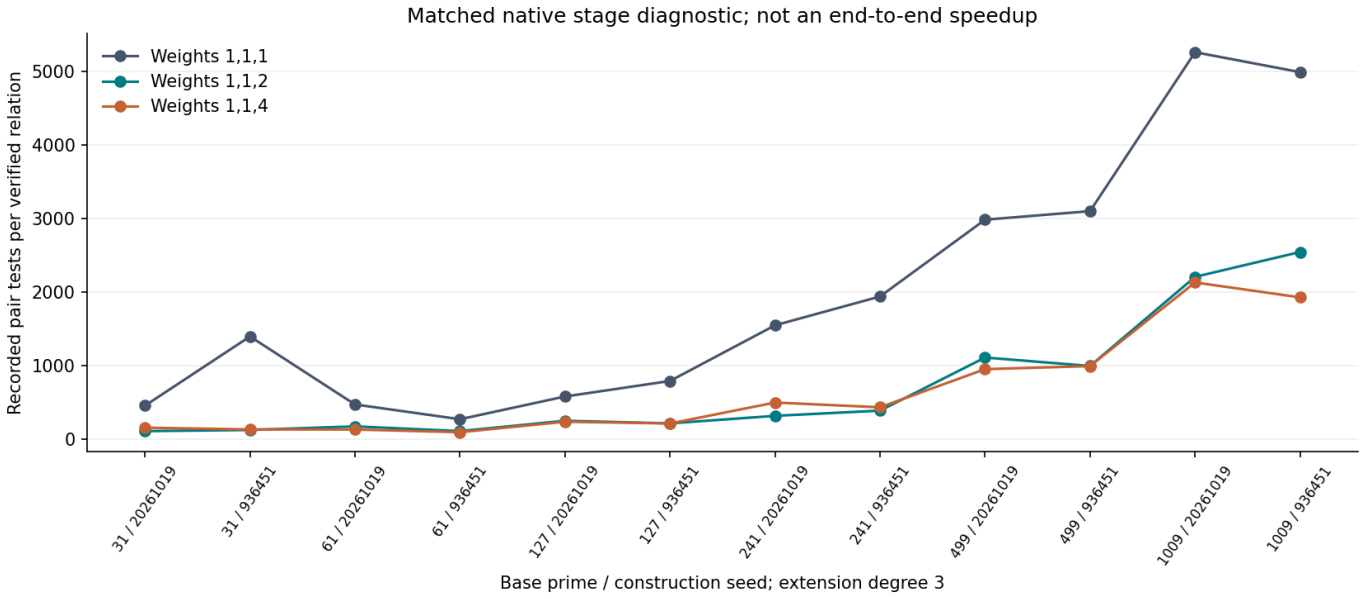


Figure 1: Recorded pair queries per verified relation; native-20261019-replay

p	construction seed	group order	unknowns	weight	verified / targets	pair tests	pair tests / relation
31	20261019	30047	11	1	6 / 128	2714	452.333
31	20261019	30047	11	2	24 / 128	2537	105.708
31	20261019	30047	11	4	17 / 128	2605	153.235
31	936451	29629	11	1	2 / 128	2788	1394.000
31	936451	29629	11	2	21 / 128	2564	122.095
31	936451	29629	11	4	20 / 128	2581	129.050
61	20261019	227893	30	1	15 / 128	7034	468.933
61	20261019	227893	30	2	38 / 128	6483	170.605
61	20261019	227893	30	4	48 / 128	6143	127.979
61	936451	227111	34	1	27 / 128	7230	267.778
61	936451	227111	34	2	60 / 128	6385	106.417
61	936451	227111	34	4	69 / 128	6313	91.493

p	construction seed	group order	unknowns	weight	verified / targets	pair tests	pair tests / relation
127	20261019	2047933	63	1	24 / 128	13849	577.042
127	20261019	2047933	63	2	52 / 128	12848	247.077
127	20261019	2047933	63	4	53 / 128	12395	233.868
127	936451	2048113	62	1	18 / 128	14197	788.722
127	936451	2048113	62	2	57 / 128	12055	211.491
127	936451	2048113	62	4	56 / 128	11768	210.143
241	20261019	13993069	130	1	19 / 128	29373	1545.947
241	20261019	13993069	130	2	71 / 128	22303	314.127
241	20261019	13993069	130	4	52 / 128	25741	495.019
241	936451	14001257	116	1	14 / 128	27131	1937.929
241	936451	14001257	116	2	58 / 128	22333	385.052
241	936451	14001257	116	4	53 / 128	22813	430.434
499	20261019	124249159	249	1	19 / 128	56670	2982.632
499	20261019	124249159	249	2	47 / 128	52049	1107.426
499	20261019	124249159	249	4	52 / 128	49357	949.173
499	936451	124261523	258	1	19 / 128	58878	3098.842
499	936451	124261523	258	2	51 / 128	50621	992.569
499	936451	124261523	258	4	52 / 128	51520	990.769
1009	20261019	1027186051	495	1	21 / 128	110487	5261.286
1009	20261019	1027186051	495	2	45 / 128	99211	2204.689
1009	20261019	1027186051	495	4	47 / 128	100052	2128.766
1009	936451	1027286801	495	1	22 / 128	109779	4989.955
1009	936451	1027286801	495	2	42 / 128	106812	2543.143
1009	936451	1027286801	495	4	52 / 128	100150	1925.962

Synthetic one-target recoveries

p	seed	weight	residuals	independent rank	status
31	20261019	1	211	12	verified_recovery
31	20261019	2	81	11	verified_recovery
31	20261019	4	66	12	verified_recovery
31	936451	1	146	11	verified_recovery
31	936451	2	64	10	verified_recovery
31	936451	4	43	12	verified_recovery
61	20261019	1	210	31	verified_recovery
61	20261019	2	75	29	verified_recovery
61	20261019	4	97	30	verified_recovery
61	936451	1	171	32	verified_recovery
61	936451	2	64	30	verified_recovery
61	936451	4	47	29	verified_recovery
127	20261019	1	431	59	verified_recovery
127	20261019	2	161	61	verified_recovery
127	20261019	4	164	63	verified_recovery
127	936451	1	526	59	verified_recovery
127	936451	2	148	56	verified_recovery
127	936451	4	126	58	verified_recovery
241	20261019	1	584	117	verified_recovery
241	20261019	2	248	119	verified_recovery
241	20261019	4	251	121	verified_recovery
241	936451	1	724	105	verified_recovery
241	936451	2	296	105	verified_recovery
241	936451	4	292	110	verified_recovery

Completed 24 recoveries; 0 budget-exhausted recoveries. Total independently checked PDP decisions, including A/A controls and recovery attempts: 11370. The independent comparison is against a separate dictionary, not an independent host.

Limits and next checks

The legacy pair API does not emit internal lookup, polynomial addition/division, allocation or RNG counts. Its F_p multiplication counter also uses legacy inverse accounting. These missing counts prevent a complete pricing band, S, or ranking against rho. No strongest-external-solver comparison was run: Magma, msolve and FGb were not run here. Base and weighted-slot setup, verification group operations, residual-generation group operations and modular elimination multiplications have separate ledgers. Dictionary construction, reference arithmetic and fixture generation are excluded validation/setup work without complete operation ledgers. No binary-field or ECC2K transfer was tested.

The next discriminating step is complete native counters plus matched ecbench one-target rho measurements, with independent-host replay and the fastest available external algebraic reference. These obligations remain open; verified recoveries alone do not discharge them.

Reproduction and provenance

See config.json, host.txt, SHA256SUMS, records.jsonl, the committed PROTOCOL.md and the native example. Seed 936451 is the holdout. Inputs and outputs are public synthetic fixtures. Histories from the earlier Python experiment remain historical and are not runtime baselines. All A/A decisions and deterministic counters were asserted equal.

Replay record

Run: native-20261019-replay. Source fc13c52e82202b44a13f9e32dca2fac2401e320e; analysis 040b8e1a. Exact command: the replay's config.json. All 12 curves, 48 PDP arms and 24 recoveries completed, with zero budget exhaustion. Its records.jsonl is byte-identical to the primary's; validation/replay.txt records the hash comparison. This is a same-host deterministic replay, not independent-host validation or additional independent sampling.

Host: AMD EPYC 9V74 KVM; four-CPU cgroup, 16 GiB; L0 with concurrent work. No wall time or uncertainty interval is admitted. The figure reproduces Rust-recorded per-curve values. See raw records, configuration and receipt.